

Percentage Change

AQA · KS3 Year 7

Name: _____ Class: _____ Date: _____

Calculate percentage increase and decrease. Find the percentage change between two values. Solve real-life problems involving percentage profit, loss and change.

VISUAL EXAMPLE



$$\frac{1}{2} = 0.5 = 50\%$$

shaded part = 1 of 2 equal parts = half

One half shown three equivalent ways.

COMMON MISTAKES

Dividing by the NEW value instead of the ORIGINAL when calculating percentage change.

Adding/subtracting the percentage of the new amount instead of the original.

Confusing percentage increase with the new total — an increase of 20% means the new value is 120%.

Forgetting to check if the change is an increase or a decrease.

METHOD

- 1 Find the percentage of the original: $(\text{percentage} \div 100) \times \text{original}$.
- 2 Percentage INCREASE: add the amount to the original.
- 3 Percentage DECREASE: subtract the amount from the original.
- 4 Percentage CHANGE: $(\text{actual change} \div \text{original}) \times 100$.
- 5 Check — state if the change is an increase or decrease.

WORKED EXAMPLE

A coat costs £80. It is reduced by 15%. Find the sale price.

Step 1 — Find 15% of £80: $15 \div 100 \times 80 = £12$

Step 2 — It's a decrease: $£80 - £12 = £68$

Answer: The sale price is £68.

1 Find:

(2 marks)

- a) 10% of £50 b) 25% of £120 c) 5% of £200 d) 1% of £350

HOW TO START

To find 10%, divide the amount by 10.

To find 1%, divide the amount by 100.

25% is a quarter, so divide by 4; 5% is half of 10%.

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

2 Increase each amount by the given percentage.

(2 marks)

a) £40 by 10% b) £60 by 25% c) £200 by 50%

HOW TO START

First find the percentage of the amount (just like Q1).
Then ADD that amount on to the original.

ANSWERS

(a) = _____
(b) = _____
(c) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

3

(2 marks)

a) £80 by 20%

b) £150 by 10%

c) £500 by 35%

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

(3 marks)

- a) How much did the price increase by?
b) What is the percentage increase?

(a) = _____

(b) = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

5 A bicycle cost £320. In a sale there is 15% off.

(3 marks)

- a) How much is the discount?
- b) What is the sale price?
- c) After the sale, the price goes back up by 15%. Is it £320 again? Show why or why not.

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

(3 marks)

Calculate the percentage increase from January to February.

Answer =

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

7

Mo buys trainers for £60 and sells them for £78.

(4 marks)

Abi buys a bag for £45 and sells it for £54.

- Calculate Mo's percentage profit.
- Calculate Abi's percentage profit.
- Who made the better deal? Explain.

ANSWERS

(a) = _____

(b) = _____

(c) =

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

8 A house was worth £200,000. Its value increased by 12% in Year 1, then decreased by 8% in Year 2.

(5 marks)

- a) What was the value after Year 1?
- b) What was the value after Year 2?
- c) What is the overall percentage change from the original price? Is it an increase or decrease?

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Find the percentage of the original value | 2 Increase: add it on | 3 Decrease: take it off | 4 Percentage change = $(\text{change} \div \text{original}) \times 100$ | 5 State increase or decrease

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

One method step I want to remember next time:

One thing I will practise before the next lesson:

Teacher key

Q1: a) £5 b) £30 c) £10 d) £3.50

Q2: a) £44 b) £75 c) £300

Q3: a) £64 b) £135 c) £325

Q4: a) £45 b) $(45 \div 450) \times 100 = 10\%$

Q5: a) 15% of 320 = £48 b) $320 - 48 = £272$ c) 15% of 272 = £40.80; $272 + 40.80 = £312.80 \neq £320$. No — the 15% increase is on a smaller amount.

Q6: Change = 50; $(50 \div 250) \times 100 = 20\%$ increase

Q7: a) $(18 \div 60) \times 100 = 30\%$ b) $(9 \div 45) \times 100 = 20\%$ c) Mo — higher percentage profit

Q8: a) $200,000 \times 1.12 = £224,000$ b) $224,000 \times 0.92 = £206,080$ c) Change = £6,080; $(6,080 \div 200,000) \times 100 = 3.04\%$ increase
